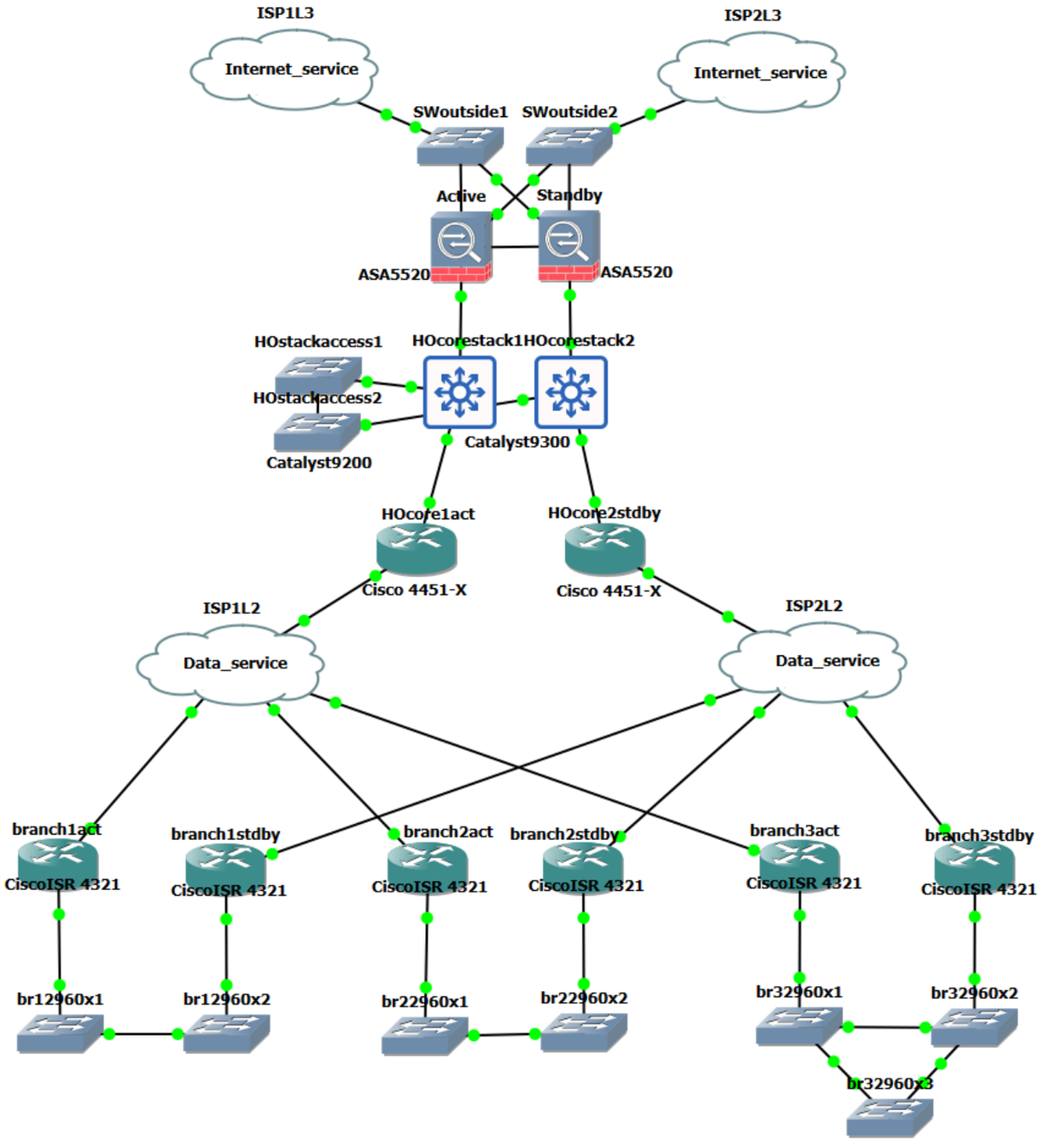
HQ and branch network topology



Design legend

Branch gateways

Each ISR 4321 pair provides an HSRP gateway. IP SLA tracks HQ reachability over the primary circuit. A failed probe lowers HSRP priority so the backup router can preempt and forward through the second provider.

WAN tunnels and EIGRP

Each branch reaches the 2 HQ routers over separate provider circuits. EIGRP runs over GRE tunnels protected by IPsec. Bandwidth and delay settings on the backup tunnel interfaces at both ends give that path a higher EIGRP metric, so both directions prefer the primary circuit during normal operation.

HQ gateways

The ISR 4451-X pair terminates the tunnels. HQ users use their VLAN's HSRP virtual IP as their default gateway. The ASA pair uses the transit VLAN's HSRP virtual IP as its next hop to internal networks.

HQ switching

The Catalyst 9200 access stack has uplinks to both members of the Catalyst 9300 core stack.

Internet edge

The ASA 5520 pair operates in active/standby mode. Each firewall connects to both Internet providers.

Branch 3 switching

The 3 standalone Catalyst 2960-X switches use per-VLAN spanning tree to prevent loops.